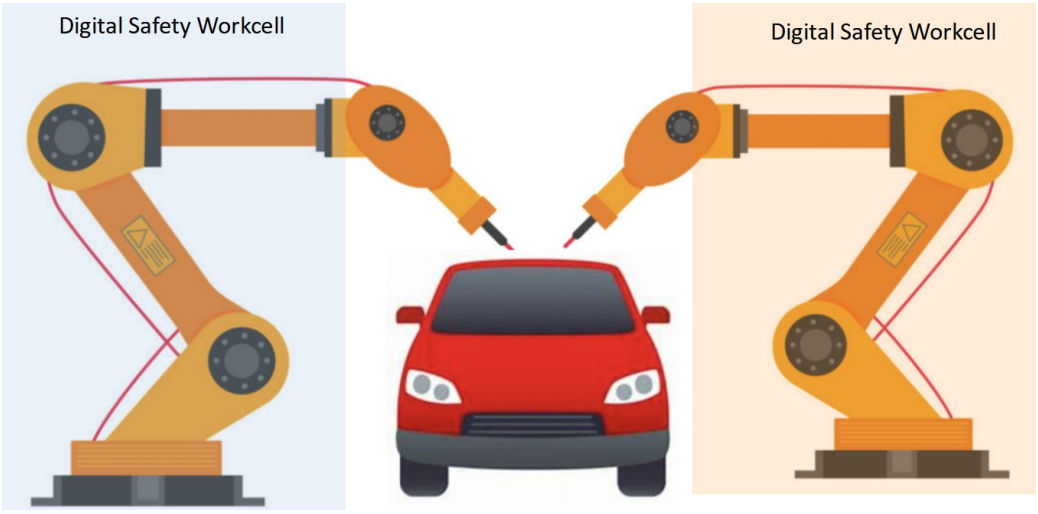
OT Digital Safety



Safely caged inside digital workcells robots make cars, medical devices and lab equipment run tests, building systems keep us comfortable, and critical infrastructure is secure. Applying the OT Zero Trust philosophy to operations segments the OT network into Digital Safety Workcells shielded by OT Firewalls and OT IPS that only allow trustworthy control commands to enter. Ransomware is stopped before it can encrypt data.

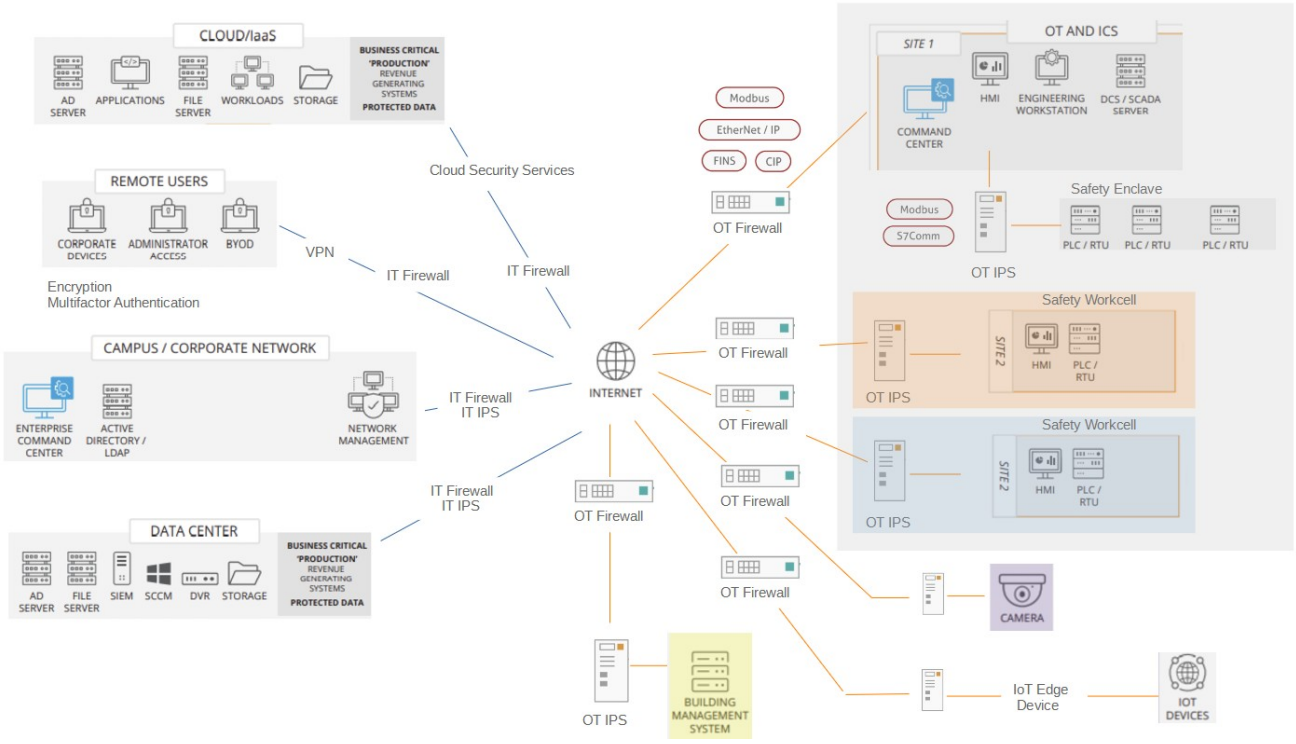
versus IT cybersecurity bolted onto OT operations



Bolting IT Zero Trust controls onto cyber physical systems grinds production schedules to a crawl or crashes systems because OT devices were not designed to decrypt commands and check multifactor authenticator apps. Malware lurks in IT network traffic. One infected machine can spread ransomware.

Deploy OT Digital Safety

This diagram shows where to deploy OT firewalls and OT IPS to segment an enterprise network. Digital Safety Workcells are shown in gray, orange, blue, yellow, and purple.



OT firewalls and OT IPS rely on threat intelligence to generate block lists. The OT defense console shows what's happening across operations in near real-time. Cyber physical devices are scanned by OT portable scanners to make sure they start work with a clean bill of health per regulations. Security updates are continual so you never trust – always verify digital safety.

Zero Trust Philosophy Demystified

Published in 2020, the NIST.SP.800-207 Zero Trust Architecture guide explains that any person, computer, or robot can become untrustworthy at any time. You cannot rely on network perimeter safeguards because cloud services, remote work, accessing the internet for maintenance updates, and other new technologies require new cyber defenses. Read: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-207.pdf It's about time. The key difference between IT and OT is the work schedule. Humans work on information technology (IT) systems on a need-to-know basis. Time is measured in minutes. The time it takes for workers to check in using multifactor authenticator (MFA) apps or keyfobs does not substantially slow down their productivity. The benefit of encrypting data outweighs the cost of the computing power necessary to do the calculations. Operations technology (OT) works to meet strict production schedules. Machines perform repetitive tasks just in time, every time, all the time. Time is measured in seconds, milliseconds, or nanoseconds. In 2022, machines don't have the time or computing power to decrypt control commands or check MFA to prove their trustworthiness. Tailor zero trust to operations. Protect machines with OT Digital Safety.